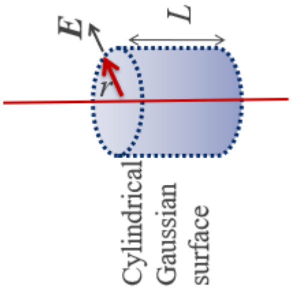
MST 2026 Detailed solution:
Question 1
(a)
(i)



1. Identify symmetry
2. What should be the Gaussian surface here?
3. Apply $\oint_s \vec{E} \bullet d\vec{s} = \frac{Q_{enclosed}}{\varepsilon}$
4. What is the $Q_{enclosed}$ in the Gaussian surface ?

$$Q_{enclosed} = \int_0^L \rho_l dl = \rho_l L$$

5. What is Gaussian surface area? $\left| \oint_s d\vec{s} \right| = 2\pi r L$
6. Substitute in GL and solve for E,

$$\oint_s \vec{E} \bullet d\vec{s} = \frac{Q_{enclosed}}{\varepsilon_0} \rightarrow 2\pi r L \left| E \right| = \frac{\rho_l L}{\varepsilon_0}$$

$$\therefore \vec{E} = \left| \frac{\rho_l}{2\pi\varepsilon_0 r} \right| \vec{a}_r \quad \text{V/m}$$

(ii)

Electrical field at radial distance r from the centre of a cylindrical density ρ_l [C/m]:

$$\boxed{\oint_s \vec{E} \bullet d\vec{s} = \frac{Q_{enclosed}}{\varepsilon_0} \rightarrow 2\pi r L \left| E \right| \vec{a}_R = \frac{\rho_l L}{\varepsilon_0}} \Rightarrow \boxed{\vec{E} = \frac{\rho_l}{2\pi\varepsilon_0 r} \vec{a}_r}$$

(like example 2)

Potential difference between two radial points:

$$V_{21} = -\int_{r_1}^{r_2} \vec{E} \bullet d\vec{l} = -\int_{r_1}^{r_2} \frac{\rho_l}{2\pi\varepsilon_o r} dr = \frac{\rho_l}{2\pi\varepsilon_o} \ln \left(\frac{r_1}{r_2} \right)$$

(iii)
The ground plane is a negatively charged body. If we consider its influence on the electric field of the charged line, the Gaussian symmetry will be lost as the field line will not remain normal on the Gaussian surface **everywhere** around the line.

(b)

$$V_{21} = \frac{\rho_l}{2\pi\varepsilon_o} \ln \left(\frac{r_1}{r_2} \right) = 12.694kV$$

(c) While calculating capacitance between the line and the ground plane, we cannot ignore the influence of the plane anymore. Hence, the derivation of potential using Gauss's law is invalid. The method of image brings an easier solution, which converts the ground plane to an equipotential surface and an image of the line below the surface. The potential at the surface is the same as the ground plane, i.e., 0, which maintains the boundary conditions of Laplace's equation. The potential on the line and the image can be found using similar triangles and superposition, which then can be used to find the capacitance.

Question 2

(a)

Polarization: dipole formation at the microscopic level

Polarization vector: dipole moments per unit volume.

(b)

$$\therefore \varepsilon_0\vec{E}+\vec{P}=\vec{D} \quad \rightarrow \quad \vec{P}=\vec{D}-\varepsilon_0\vec{E}=\varepsilon_0(\varepsilon_r-1)\vec{E}, \quad \vec{D}=\varepsilon_0\varepsilon_r\vec{E}$$

$$E=\frac{Q}{\varepsilon_0\varepsilon_r S}$$

$$P=\varepsilon_0(\varepsilon_r-1)\frac{Q}{\varepsilon_0\varepsilon_r S}=\frac{(\varepsilon_r-1)Q}{\varepsilon_r S}=8.89\times10^{-7}\,C/m^2$$

(c)

$$|D_{1n}|=|D_{2n}| \;\rightarrow\; \varepsilon_1|E_{1n}|=\varepsilon_2|E_{2n}|$$

$$|E_{1n}|=\frac{\varepsilon_2}{\varepsilon_1}|E_{2n}|=\frac{6}{3}E_{2n}=2E_{2n}$$

$$|E_{2n}|=\frac{1}{6\varepsilon_0}\frac{Q}{S}$$

Question 3

$$\begin{bmatrix} Q_1 \\ Q_2 \\ Q_3 \end{bmatrix}=\begin{pmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{pmatrix}\begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix}=\begin{pmatrix} (C_{10}+C_{12}+C_{13}) & -C_{12} & -C_{13} \\ -C_{12} & (C_{20}+C_{12}+C_{23}) & -C_{23} \\ -C_{13} & -C_{23} & (C_{30}+C_{13}+C_{23}) \end{pmatrix}\begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix}$$

$$C=\begin{pmatrix} 20 & -5 & -5 \\ -5 & 20 & -5 \\ -5 & -5 & 20 \end{pmatrix}$$

Question 4

An isolated magnetic pole cannot exist, i.e., there is no divergence of **B**.

Magnetic flux lines always close upon themselves or circulate around the current that is producing it, i.e., the curl of **B** is non-zero but equal to current density J and permeability of the medium.